

Simple 10-Step Kubernetes Deployment List

1. **Containerize Your App:** Package your application code and dependencies into a Docker image.
2. **Push to Registry:** Upload your image to a container registry like Docker Hub, Amazon ECR, or Google Artifact Registry.
3. **Prepare Your Cluster:** Ensure you have access to a running Kubernetes cluster (local like Minikube or cloud-based like EKS/GKE).
4. **Write Deployment YAML:** Create a manifest file defining your desired state (image name, replicas, and resource limits).
5. **Define a Service:** Create a Service YAML to provide a stable network identity and load balancing for your pods.
6. **Configure Environment:** Use ConfigMaps for settings and Secrets for sensitive data like API keys or passwords.
7. **Apply Manifests:** Run `kubectl apply -f <filename>` to send your configurations to the Kubernetes API server.
8. **Verify Rollout:** Check the status using `kubectl get pods` and `kubectl rollout status` to ensure everything is running correctly.
9. **Expose Externally:** Set up an Ingress controller or a LoadBalancer service to make your app accessible to the internet.
10. **Enable Scaling:** Configure a Horizontal Pod Autoscaler (HPA) to automatically adjust replicas based on traffic demand.